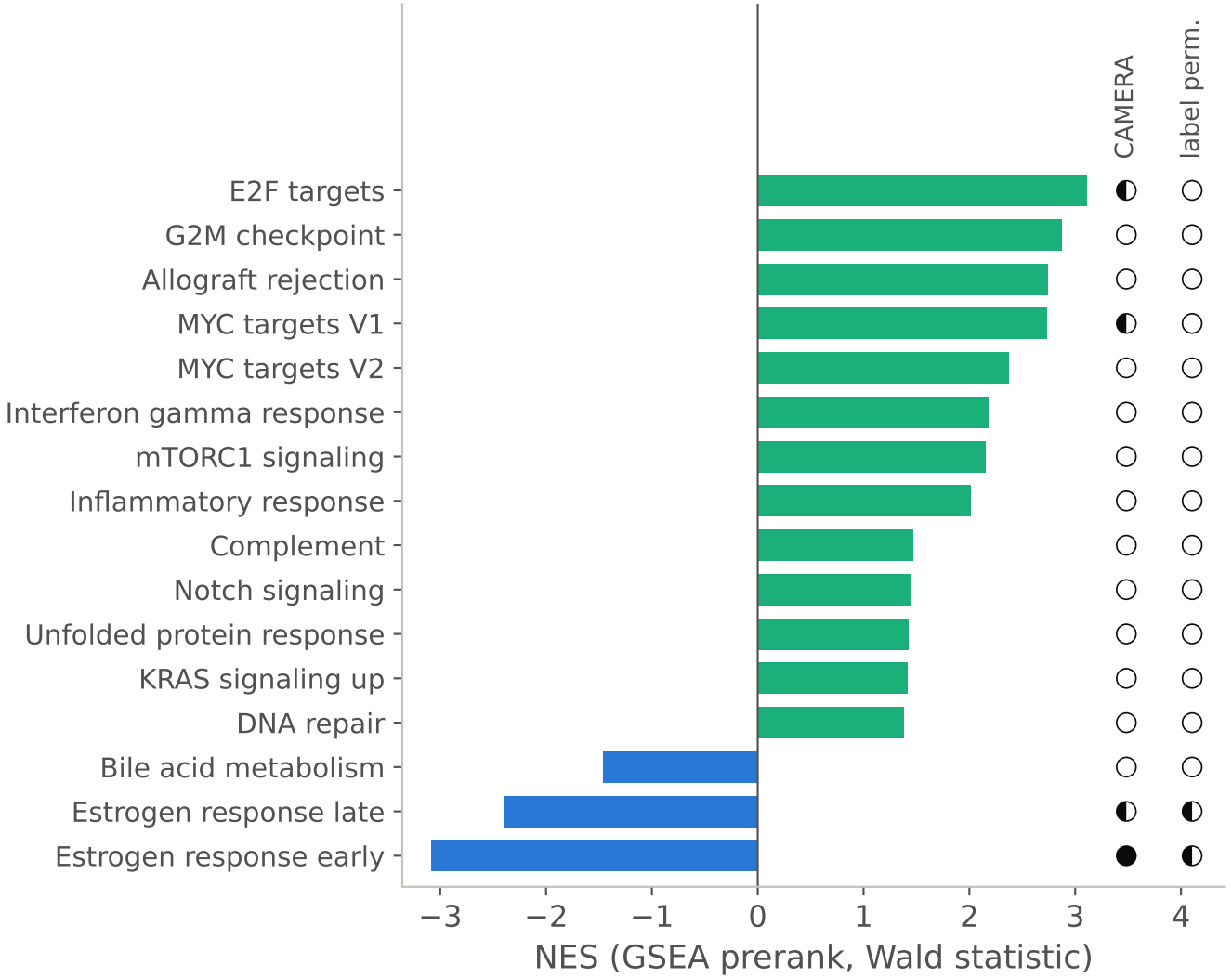
TNBC ranks higher on proliferation, MYC and immune programmes, lower on estrogen response.

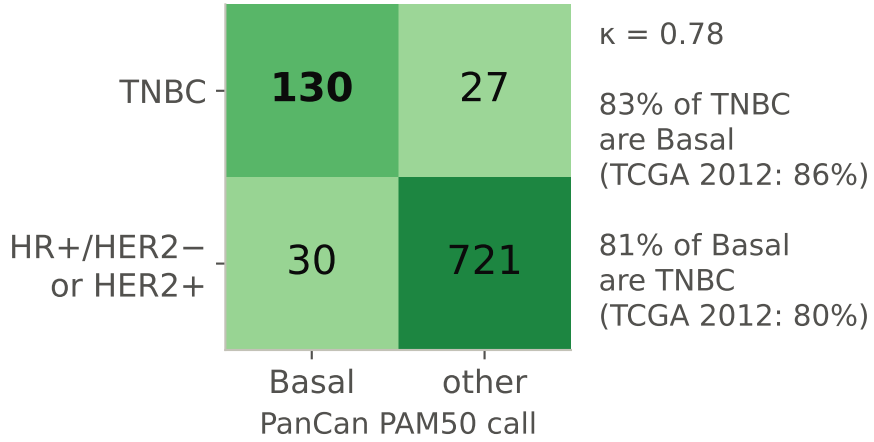
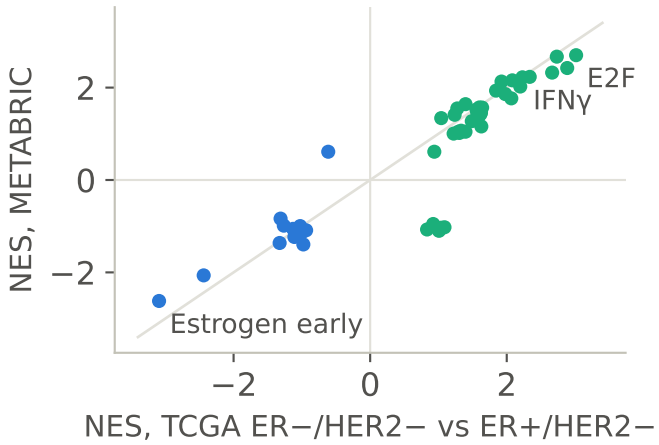
Correlation-aware checks agree on direction; only early estrogen response stays significant.

NES: GSEA prerank on the DESeq2 Wald statistic (gene-permutation q-values are optimistic when genes are co-expressed).
Marks: ● q < 0.05, ◐ nominal p < 0.05, ○ neither, for CAMERA (competitive test with inter-gene correlation) and label-permutation GSEA.

Hallmarks at prerank FDR < 0.05 (16 of 28 shown)



Clinical TNBC vs PAM50 Basal



METABRIC replicates: NES ρ = 0.94